AIMO Proof Pilot — Final Grade

Problem (#_ID): 2_ALTORO **Submission:** model_5

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

1. Final mark

Final mark (choose one): $\boxed{6}$

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. `ME1`) where possible.

An essentially complete solution. It sets up coordinates, applies Menelaus to obtain X, computes HQ, shows P is the midpoint of HQ, introduces the reflection T of H in AC with $PQ/PE = HQ/QT$, and correctly computes the area of ABC and the circumradius. Two justifications are thin: the computation of $\sin \angle QAT$ is correct but the formula used (a vector cross product) is not derived, and the coordinates of A are only lightly justified. Crucially, unlike model 6, the actual calculations here are all present and checkable, so these gaps amount to a minor error rather than a fundamental omission and do not drag the mark down to 1. Essentially complete with a minor error, so scores **6**.